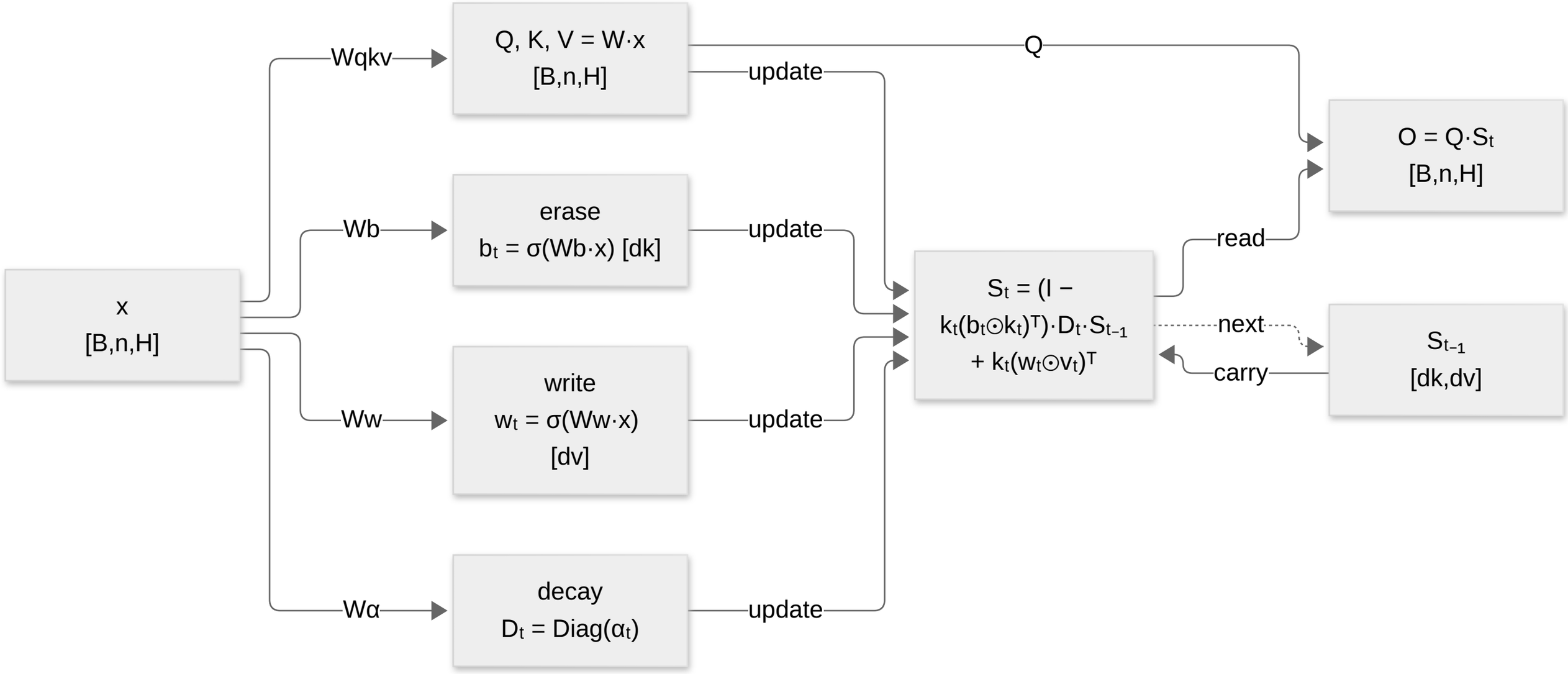

x
[B,n,H]
Wqkv
Wb
Ww
Wα
Q, K, V = W·x
[B,n,H]
erase
b_t = σ(Wb·x) [dk]
write
w_t = σ(Ww·x)
[dv]
decay
D_t = Diag($α_t$)
update
update
update
update
Q
S_t = (I − $k_t(b_t⊙k_t)^T)·D_t·S_{t-1}$ + $k_t(w_t⊙v_t)^T$
read
next
carry
O = Q·S_t
[B,n,H]
S_{t-1}
[dk,dv]